



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY

LONERE – RAIGAD – 402 103

Winter Semester Examination – December-2019

Branch: B. Pharm.

Subject: Medicinal Chemistry-I (BP402T)

Date: 06/12/2019

Semester: IV

Marks: 75

Time: 3hrs

- Instructions:**
- i) All questions are compulsory
 - ii) Figures to the right indicate full marks
 - iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Multiple Choice Questions (MCQs)

(20 Marks)

1. Aspirin introduced by -----in 1889
 - A. Dreser's.
 - B. Emil Fischer and Mering.
 - C. Barger and Dale.
 - D. Non of Above.
2. Bioisosteres are similar in their-
 - A. Physical Character.
 - B. Chemical Character.
 - C. Both Character a & b.
 - D. Biochemical Character.
3. Which of the following reaction is not a Phase I metabolic transformation?
 - A. Reduction of ketones.
 - B. Conjugation of alcohols.
 - C. Monoamine oxidases.
 - D. Esterhydrolysis.
4. -----reaction is most common in drugs having ester or amide functional groups
 - A. Oxidative reactions.
 - B. Reduction.
 - C. Hydrolysis.
 - D. Phase I.
5. Which of the following adrenergic drug belongs to category of vasoconstrictors?
 - A. Metaraminol.
 - B. Isoprenaline.
 - C. Amrinone.
 - D. Naphazoline.
6. Which type of Receptor in Bronchial Smooth Muscle.
 - A. Alpha 1 subtype.
 - B. Beta 1 subtype.
 - C. Alpha 2 subtype
 - D. Beta 2 subtype.
7. Prazocin belongs to the class of
 - A. Pyridyl quinoxaline.
 - B. Piprazinyl quinoxaline.
 - C. Pyridyl quinazoline
 - D. Piprazinyl quinoxaline
8. In Propranolol the aromatic ring and its substitute is the primary determining of which activity.
 - A. Alpha 1 antagonistic.
 - B. Alpha 2 antagonistic.
 - C. Beta 1 antagonistic.
 - D. Beta 2 antagonistic.
9. Ach agonist drug having IUPAC 2(Acetoxy)-N, N, N-trimethyl propan-1-aminium
 - A. Methacholine.
 - B. Bethanechol.
 - C. Carbachol.
 - D. Pilocarpine.

10. IUPAC name of Pralidoxime chloride is-
- 2-[(Hydroxyamino)methyl]-1-methyl pyridine-1-ium chloride.
 - Ethyl-(3-hydroxyhynyl)dimethyl-ammonium chloride.
 - 3-[(Dimethylamino carbonyl)oxy]-1-methyl pyridinium chloride.
 - 2 (Acetoxy)-N, N, N-trimethyl propan-1-aminium chloride.
11. Tropicamide contain which amine group-
- Primary amine.
 - Secondary amine.
 - Tertiary amine.
 - Contain no amine group.
12. Which replacement is general structure of anti-cholinergic drug make it useful in case of peptide ulcer patient?
- Replacement of N from tertiary to quaternary.
 - Change in R1, R2, R3.
 - Length of alkyl chain.
 - None of the above.
13. One of the following is not a triazolobenzodiazepine derivative
- Alprazolam.
 - Triazolam.
 - Midazolam.
 - Estazolam.
14. Blood levels of neuroleptics are useful to check the following:
- Toxicability.
 - Therapeutic levels.
 - Resistance.
 - All of the above.
15. The Mechanism of action of Carbamazepine appears to be similar to that of:
- Benzodiazepines.
 - Sodium valproate.
 - Phenytoin.
 - Ethosuximide.
16. The anesthetic agent, which is an inhibitor of NMDA glutamate receptors is:
- Thiopental.
 - Halothane.
 - Ketamine.
 - Sevoflurane.
17. Which COX enzyme are aspirin, indomethacin, and sulindac more selective for?
- COX-1.
 - COX-2.
 - They are equally selective for COX-1 & COX-2.
18. Most current opioid analgesics act at this opioid receptor subtype
- Kappa.
 - Delta.
 - Epsilon
 - Mu.
19. Drug associated with the hepatic/renal toxic metabolite: N-acetyl-p-benzoquinone.
- Diclofenac.
 - Meclofenamate.
 - Indomethacin.
 - Acetaminophen.
20. What can be done to prevent GI side effect of NSAIDs?
- Co-administration of NSAID with PPI.
 - Enteric coated tablets.
 - Prescribe CINODs instead.
 - All of the above.

Q. No.2 Long Answers (Answer any two)

(20 Marks)

- A] Classify Antipsychotics with example & Give SAR of any three categories. Sketch out Synthetic pathway of phenytoin.
- B] Define Drug metabolism. Write about Phase-I & Phase-II reactions.
- C] Give the classification of Narcotic analgesics. Discuss SAR & MOA of Morphine Analogue. Draw schematic pathway for synthesis of methadone.

Q. No.3 Short Answers (Answer any Seven)

(35Marks)

- A] Explain classification of bioisosterism.
- B] Elaborate Factors affecting drug metabolism.
- C] Write few example of mixed action of sympathetic drug with IUPAC & structure.
- D] Highlights two example of Nonselective alpha adrenergic blocker with MOA & Structure.
- E] What are cholinergic receptor? Classify them. Sketch out synthetic pathway of Halothane & Ketamine.
- F] Classify anticonvulsant with example. Discuss SAR of Barbiturates.
- G] Give SAR & MOA of Fenamates.
- H] Explain biosynthesis of & Catabolism of catecholamine.
- I] What is mean by narcotic antagonist. Draw structure & Therapeutic use of Naloxon & Levallorphan.

----- END OF PAPER -----



- a) Heart
c) Adipose tissue
viii) Muscarinic receptors are _____ receptor.
a) Nuclear
c) Enzyme
ix) Which of the following drug is a direct acting sympathomimetic drug.
a) Phenylephrine
c) Propylhexedrine
x) Which of the drug is alpha adrenergic blocker?
a) Propranolol
c) Atenolol
xi) Acetylcholine on hydrolysis gives....
a) Serine and acetic acid
c) Acetic acid and serine
xii) Chlordiazepoxide, a psychotherapeutic agent belongs to class of _____.
a) Carbamates
c) Benzodiazepine
xiii) Acetylcholine is biosynthesized from _____.
a) L-Cysteine
c) L-Serine
xiv) Hypnotics are _____.
a) Strong depressant of CNS
c) Mild depressant of CNS
xv) Chemical name 5-ethyl, 5-phenyl barbituric acid belong to
a) Hexobarbital
c) Pentobarbital
xvi) Which one of the following receptor is block by Neuroleptic agent?
a) Cholinergic
c) GABA
xvii) Which of the Anticholinergic agent useful as a spasmolytic drug?
a) Pyridostigmine
- b) Bronchi
d) Skeletal muscles.
b) G-protein Coupled
d) Ligand gated Ion Chanel.
b) Pseudoephedrine
d) Ephedrine
b) Metibranolol
d) Tolazoline
b) Choline and carbamic acid
d) Choline and acetic acid
b) Propanediol
d) Phenothiazine.
b) L-Codeine
d) L-Cholic acid
b) Strong stimulant of CNS
d) Mild stimulant of CNS
b) Phenobarbital
d) Secobarbital
b) Adrenergic
d) Dopaminergic
b) Dicyclomine

c) Tropicamide

d) Glycopyrrolate

xviii) Phenobarbital is orally administered in the treatment of:

a) Grand mal epilepsy

b) Petit mal epilepsy

c) Jackson epilepsy

d) Psychomotor epilepsy.

xix) Which of the following barbiturate derivative acts as anticonvulsant agent?

a) Hexobarbital

b) Phenobarbital

c) Pentobarbital

d) Secobarbital

xx) Which of the following drugs is acts as Cholinergic Blocking agent?

a) Physostigmine

b) Neostigmine

c) Pyridostigmine

d) Dicyclomine

Q. 2. Long Answers = $2 \times 10 = 20$ (Answer 2 out of 3)

i) Discuss the various physicochemical properties affecting the biological activity of drugs.

ii) Classify sedatives and hypnotics. Add a note on SAR and MOA of Barbiturates.

iii) Discuss the SAR of Parasympathomimetic agents and add a note on cholinergic receptors.

Q. 3. Short Answers = $7 \times 5 = 35$ (Answer 7 out of 9)

i) Discuss SAR of sympathomimetic drugs. Add a note on direct acting sympathomimetic drugs.

ii) Describe SAR of Morphine analogues. Add a note on Narcotic antagonists.

iii) Describe the factors that affect drug metabolism.

iv) Define Epilepsy and describe general mechanism of action of anticonvulsant drugs.

v) Classify anticonvulsant drugs and add a note on hydantoin derivatives.

vi) Classify antipsychotic drugs. Add a note on SAR and MOA of phenothiazine derivatives.

vii) Write a note on Biosynthesis and catabolism of catecholamine.

viii) Outline the synthesis of Phenytoin and Barbitol.

ix) Write a note on adrenergic receptor. Classify sympathomimetic drugs with examples

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 09/02/2023

Course : B. Pharmacy
Subject Name : Medicinal Chemistry-I
Max Marks : 75

Sem: IV
Subject Code: BP402T
Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Discuss the importance of partition coefficient in relation to biological activity.
- ii) What are neurotransmitters? Classify them with suitable examples.
- iii) Outline the chemical classification of adrenergic drugs with suitable examples.
- iv) Write the MOA of NSAIDs.
- v) Give the biosynthetic pathway of Acetylcholine.
- vi) Draw structure and IUPAC name of Piroxicam.
- vii) What do you mean by tonic clonic seizures?
- viii) Discuss the factors Stereo chemical aspects in relation to drug metabolism.
- ix) Define and distribute adrenergic receptors.
- x) Differentiate generalized seizures & partial seizures

Q. 2. Long Answers (Answer 2 out of 3)

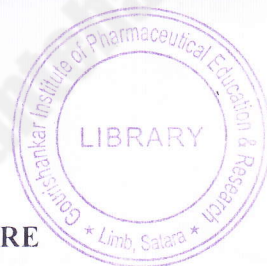
(2 x 10) = 20

- i) Define & classify para sympathomimetic agents with suitable example. Give the SAR, mode of action & chemical features of each class.
- ii) Outline the chemical classification of adrenergic drugs. Discuss their mechanism of action. Comment on the essential structural features required for the optimum activity of such drugs.
- iii) Explain in detailed history and development of medicinal chemistry.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Outline the synthesis of diazepam and chlorpromazine hydrochloride
- ii) Classify psychotherapeutic drugs with suitable example. . Discuss the SAR of phenothiazine's as tranquilizing agents.
- iii) Discuss the influence of bioisosterism and partition coefficient on biological activity.
- iv) What are synthetic analgesics? Explain the synthesis and uses of methadone hydrochloride.
- v) Define & classify general anesthetics. Write in brief steps involved in general anesthesia.
- vi) Write a note on biosynthesis and catabolism of catecholamine.
- vii) Discuss on different beta receptor antagonists and write the limitations of non- selective beta blockers.
- viii) What are synthetic analgesics? Explain the synthesis and uses of methadone hydrochloride.
- ix) What are anticholinesterases? Classify & describe the chemistry, MOA & therapeutic uses of the same.



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Course : B. Pharmacy
Subject Name : Medicinal Chemistry-I

Date: 09/02/2023
Sem: IV
Subject Code: 402T
BP402T
Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. One or Two answer sentence (Answer all the questions) (10 x 2) = 20

- i) Describe in detail about Bioisosterism.
- ii) Write down the synthesis of Tolazoline.
- iii) Explain β adrenoreceptors.
- iv) Draw the structure and IUPAC name of Tacarine hydrochloride
- v) How does benzodiazepine affect the action of GABA?
- vi) Write MOA of major tranquillizer
- vii) Why phenytoin does not give sedation through rest drugs of hydantoin class shows sedation
- viii) Write a note on Opioid antagonist.
- ix) Write a MOA of Aspirin
- x) What is partition coefficient?

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Discuss the SAR of Sympathomimetic agent and write a note on Biosynthesis & Catabolism of Catecholamine.
- ii) Classify narcotics analgesic. Add a note on SAR and MOA Fentanyl citrate and Methadone hydrochloride.
- iii) Enlist various physicochemical properties affecting the biological activity of drug. Add a note on Phase-I reaction.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

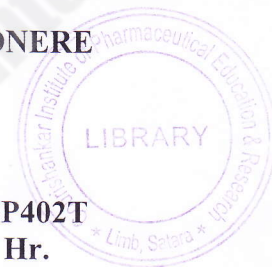
- i) What is mean by drug metabolism? Describe the factors affecting drug metabolism.
- ii) Write a note on biosynthesis and release of acetylcholine.
- iii) Outline the synthesis of Salbutamol and Ibuprofen.

- iv) Classify sedative and hypnotics. Write the SAR and MOA of Barbiturates.
- v) Write a note on cholinergic blocking agent. And write the SAR of Ipratropium bromide.
- vi) Classify antipsychotic drug. Add a note on Chlorpromazine hydrochloride.
- vii Define epilepsy and describe general mechanism of action of anticonvulsant drug.
- viii Write a note on adrenergic receptor. Classify sympathomimetic drugs with e.g.
- ix) Describe the SAR of morphine analogues. Add a note on anti-inflammatory agent.

-----END OF THE PAPER-----

Course : B. Pharmacy
Subject Name : Medicinal chemistry I
Max Marks : 75

Sem: IV
Subject Code : BP402T
Duration : 3 Hr.

**Instructions:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Write about Partition coefficient in relation to biological action.
- ii) Define bio-isosters and classify with examples.
- iii) Draw structure and write mechanism of action and medicinal uses of Phenylephrine.
- iv) What are cholinolytic agents write mechanism of action with example.
- v) Write the MOA of Neostigmine.
- vi) What are adrenergic receptors? How will you classify them?
- vii) Outline the synthesis of Salbutamol.
- viii) Discuss alpha adrenergic antagonists with examples.
- ix) Write mechanism of action with example of solanaceous alkaloids and analogs.
- x) Draw structure and write mechanism of action of Chlorpromazine hydrochloride.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) What is epilepsy? Classify anticonvulsants with examples. Discuss Hydantoins as anticonvulsants and outline synthesis of Phenytoin.
- ii) Elaborate on biosynthesis, catabolism of Acetylcholine. Draw synthesis and write mechanism of action of carbachol.
- iii) What are catecholamines? Outline biosynthetic and metabolism pathway of catecholamines in body.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain with examples Cholinesterase enzyme inhibitors.
- ii) Write a note on importance of stereo chemistry in drug action.
- iii) What are different factors affecting drug metabolism.
- iv) Write in detail about SAR and mechanism of Phenothiazine as Antipsychotics with examples.
- v) What are sedatives and hypnotics? Write SAR of Benzodiazepines.
- vi) Explain the Mechanism of action and classification of General anaesthetics and draw synthesis of Halothane.
- vii) Write a note on Beta blockers with examples.
- viii) Explain in detail chemistry and Mechanism of action of non narcotic anti-inflammatory agent with examples.
- ix) Explain classification of opioid analgesics. Explain SAR and therapeutic uses of Mepiridine analogs.



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: IV

Subject Name: Medicinal Chemistry I

Subject Code: BP402T

Max Marks: 75

Date: 15-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10X2) =20

- i) Classify General anaesthetics with examples.
- ii) Write a note on Cholinesterase reactivator.
- iii) Describe effect of Ionization on drug action.
- iv) Discuss alpha adrenergic antagonists with examples.
- v) Write about partition coefficient in relation to biological action.
- vi) Explain mechanism of action of non-narcotic anti-inflammatory agent with example.
- vii) Give structure and mechanism of action of Neostigmine.
- viii) Define Bio-isosters and classify with examples.
- ix) Outline the synthesis of salbutamol
- x) What are cholinergic receptors? How will you classify them.

Q.2. Long Answers (Answer 2 out of 3) (10X2) =20

- i) What are antipsychotic agents? Write classification of antipsychotic agents with examples. Write SAR of Phenothiazines as antipsychotic agents.
- ii) What are sympathomimetics? Elaborate on biosynthesis and metabolism pathway of Noradrenaline.
- iii) What are Narcotic analgesics? Write a note on modifications of Morphine nucleus.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Give SAR of Beta adrenergic blockers and synthesis of Propranolol.
- ii) Write a note on Phase II Biotransformation.
- iii) Give SAR of Cholinergic agonists.
- iv) What are different factors affecting drug metabolism.
- v) Write a detailed note on AChE inhibitors.
- vi) Write SAR of Benzodiazepines as sedative and hypnotic agents.
- vii) Explain Narcotic antagonists in detail.
- viii) Classify anticonvulsants with examples and discuss Hydantoins as anticonvulsants.
- ix) Classify Cholinergic blocking agents and give synthesis of dicyclomine hydrochloride.

***** END OF THE PAPER *****